Introduction to Computer Control Systems, 5 credits, 1RT485

Date: 2025-03-19

Teacher on duty: Daniel Arnström

Number of problems: 5

Allowed aid: A calculator and mathematical handbooks

Grade thresholds:

grade 3	15 points
grade 4	21 points
grade 5	26 points

Some general instructions and information:

- Your solutions can be given in Swedish or in English.
- Write only on one side of the paper.
- Write your exam code and page number on all pages.
- Do not use a red pen.
- Use separate sheets of paper for the different problems (i.e. the numbered problems, 1–5).

With the exception of Problem 1, ***all your answers must be clearly motivated!*** *A correct answer without a proper motivation will score zero points!*

Best of luck!

Useful results

Laplace transform table

Table 1: Basic Laplace transforms

$f(t)$	$F(s)$	$f(t)$	$F(s)$
unit impulse $\delta(t)$	1	$\sinh(bt)$	$\frac{b}{s^2-b^2}$
unit step $1(t)$	$\frac{1}{s}$	$\cosh(bt)$	$\frac{s}{s^2-b^2}$
t	$\frac{1}{s^2}$	$\frac{1}{2b}t\sin(bt)$	$\frac{s}{(s^2+b^2)^2}$
t^n	$\frac{n!}{s^{n+1}}$	$t\cos(bt)$	$\frac{s^2-b^2}{(s^2+b^2)^2}$
e^{-at}	$\frac{1}{s+a}$	$\frac{\cos(bt)-\cos(at)}{a^2-b^2}; (a^2 \neq b^2)$	$\frac{s}{(s^2+a^2)(s^2+b^2)}$
$\frac{1}{a}(1-e^{-at})$	$\frac{1}{s(s+a)}$	$\frac{\sin(at)+at\cos(at)}{2a}$	$\frac{s^2}{(s^2+a^2)^2}$
$\frac{1}{(n-1)!}t^{n-1}e^{-at}; (n=1,2,3...)$	$\frac{1}{(s+a)^n}$		
$\sin(bt)$	$\frac{b}{s^2+b^2}$		
$\cos(bt)$	$\frac{s}{s^2+b^2}$		
$e^{-at}\sin(bt)$	$\frac{b}{(s+a)^2+b^2}$		
$e^{-at}\cos(bt)$	$\frac{s+a}{(s+a)^2+b^2}$		

Table 2: Properties of Laplace Transforms

$\mathcal{L}[af(t)] = aF(s)$	$\mathcal{L}[tf(t)] = -\frac{dF(s)}{ds}$
$\mathcal{L}[f_1(t)+f_2(t)] = F_1(s)+F_2(s)$	$\mathcal{L}\left[t^2 f(t)\right] = \frac{d^2}{ds^2}F(s)$
$\mathcal{L}\left[\frac{d}{dt}f(t)\right] = sF(s)-f(0)$	$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}F(s), \quad n=1,2,3,\ldots$
$\mathcal{L}\left[\frac{d^2}{dt^2}f(t)\right] = s^2F(s)-sf(0)-f'(0)$	$\mathcal{L}\left[f\left(\frac{t}{a}\right)\right] = aF(as)$
$\mathcal{L}\left[\int f(t)\,dt\right] = \frac{F(s)}{s}+\frac{1}{s}\left[\int f(t)\,dt\right]_{t=0}$	$\mathcal{L}\left[\int_0^t f_1(t-\tau)f(t)d\tau\right] = F_1(s)F_2(s)$
$\mathcal{L}[f(t-a)] = e^{-as}F(s)$	$\mathcal{L}[e^{-at}f(t)] = F(s+a)$

Matrix exponential

$$\boxed{e^{At} \triangleq \mathcal{L}^{-1}\left\{(sI-A)^{-1}\right\}}$$

Open-loop and sensitivity functions

$$G_o(s) = G(s)F_y(s), \qquad S(s) = \frac{1}{1+G_o(s)}, \qquad T(s) = 1-S(s)$$

State-space forms and transfer function relations

- State-space form and transfer function

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned} \quad \Rightarrow \quad \boxed{G(s) = C(sI - A)^{-1}B + D}$$

- Associated matrices

$$\mathcal{S} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} \qquad \mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

- LTI system with transfer function

$$\boxed{G(s) = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_n}{s^n + a_1 s^{n-1} + \cdots + a_n}}$$

  i) Observable canonical form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 \\ -a_2 & 0 & 1 & \cdots & 0 \\ -a_3 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -a_n & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ b_3 - a_3 b_0 \\ \vdots \\ b_n - a_n b_0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

  ii) Controllable canonical form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -a_1 & -a_2 & -a_3 & \cdots & -a_n \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & 0 & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} b_1 - a_1 b_0 & b_2 - a_2 b_0 & \cdots & b_n - a_n b_0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

- Solution to state-space equation

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x(0) = x_0$$

  can be written as

$$\boxed{x(t) = e^{At} x_0 + \int_0^t e^{A\tau} B u(t - \tau) d\tau}$$

- Observer system

$$\boxed{\dot{\hat{x}} = A\hat{x} + Bu + K(y - C\hat{x})}$$

Feedback control structures

General linear feedback in Laplace form:

$$\boxed{U(s) = F_r(s)R(s) - F_y(s)Y(s)}$$

Common control structures in this form.

- PID controller:

$$F_y(s) = F_r(s) = F(s) = K_p + \frac{K_i}{s} + K_d s,$$

  where $K_p, K_i, K_d \geq 0$

- Lead-lag controller:

$$F_y(s) = F_r(s) = F(s) = K\left(\frac{\tau_D s + 1}{\beta\tau_D s + 1}\right)\left(\frac{\tau_I s + 1}{\tau_I s + \gamma}\right),$$

  where $K, \tau_D, \tau_I > 0$ and $0 \leq \beta, \gamma < 1$

- State-feedback controller with observer:

$$\begin{aligned} F_r(s) &= \left(1 - L(sI - A + KC + BL)^{-1}B\right)\ell_0 \\ F_y(s) &= L(sI - A + KC + BL)^{-1}K \end{aligned}$$

Discrete-time state-space forms

A continuous time system with zero-order-hold input signal and sample period T can be written in discrete-time as:

$$\boxed{\begin{aligned} x(k+1) &= Fx(k) + Gu(k) \\ y(k) &= Hx(k) \end{aligned}}$$

where

$$\begin{aligned} F &= e^{AT} \\ G &= \int_{\tau=0}^{T} e^{A\tau} d\tau B = \lceil \text{if } A^{-1} \text{ exists} \rceil = A^{-1}(e^{AT} - I)B \\ H &= C \end{aligned}$$

Problem 1: basic questions (6/30)

Answer only *'true'* or *'false'*. Each correct answer gives 1 point, each wrong answer gives -1 point. Minimum total points for Part A, B, and C is 0 , respectively.

Part A

Note: Write 'skip' if your home assignment 1 score ≥ 6.5

i) The system

$$G(s) = \frac{(s+5)(s+3)}{(s^5 + s^4 + 3s^3 - 2s^2 + 1)(s-4)}$$

is input-output stable.

ii) The I-part of a PID-regulator is mainly used to improve the speed of the closed-loop system.

iii) We control the system

$$G(s) = \frac{1}{s+2}$$

with a P-controller with gain $K = 3$. The stationary error of the closed-loop system is zero.

(3 p)

Part B

Note: Write 'skip' if your home assignment 2 score ≥ 6.5

i) As the phase margin for the open-loop decreases, the closed-loop system typically becomes more oscillatory.

ii) An observable state-space system is always asymptotically stable.

iii) The following state-space system is controllable

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

(2 p)

Part C

Note: Write 'skip' if you attended/completed the fourth computer session (D4)

i) Discretizing a controller with a faster sampling time will lead to more phase margin loss compared with a slower sampling time.

(1 p)

Proposed solution to problem 1

Part A

i) False

ii) False

iii) False

Part B

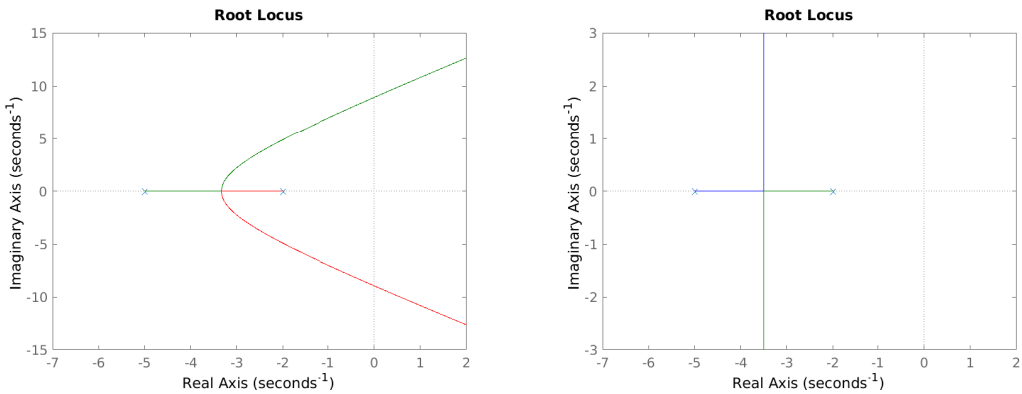
i) True

ii) False

iii) True

Part B

i) False



Root locus 1 **Root locus 2**

Figure 1

Problem 2 (6/30)

Consider a system with input $u(t)$ and output $y(t)$ which is model by

$$Y(s) = G(s)U(s)$$

with the transfer function

$$G(s) = 1/(s^2 + 6s + 5).$$

a) Assume the system if controlled with a P-controller with transfer function $F(s) = K$, for some $K > 0$. That is, $u(t) = Ke(t)$, where $e(t) = r(t) - y(t)$. Compute the closed-loop transfer function.

(2 p)

b) For the same P-controller as in a), what is the steady state error when the reference value $r(t)$ is a unit step?

(2 p)

c) Figure 1 shows two root loci. Which of the two root loci corresponds to the closed-loop system considered in a) and b)?

(2 p)

Proposed solution to problem 2

a) The closed-loop transfer function is $G_c(s) = \frac{G(s)F(s)}{1+G(s)F(s)}$. Inserting the G and F given in the exercise gives

$$G_c(s) = \frac{\frac{1}{s^2+6s+5}K}{1+\frac{1}{s^2+6s+5}K} = \frac{K}{s^2+6s+5+K}$$

(2 p)

b) Using the final value theorem gives

$$\begin{aligned}
\lim_{t\to\infty} e(t) &= 1 - \lim_{t\to\infty} y(t) \\
&= 1 - \lim_{s\to 0} sY(s) \\
&= 1 - \lim_{s\to 0} sG_c(s)\frac{1}{s} \\
&= 1 - G_c(0) \\
&= 1 - \frac{K}{5+K} \\
&= \frac{5}{5+K}.
\end{aligned}$$

(2 p)

c) The poles for the closed-loop system are solution to the equation

$$s^2 + 6s + 5 + K = 0 \Leftrightarrow (s+3)^2 - 4 + K.$$

Hence, for increasing K, the poles will initially approach -3, and then split off into complex poles when $K = 4$. The real part will then be constantly -3. The general appearance of the root locus will, hence, be like Root locus 2. However, the value of the starting points (which should be the poles for the closed-loop when $K = 0$, i.e., -1 and -5) and the "split-off point" (which should be -3) are different. Hence, the root locus for the closed-loop system does not match Root locus 1 nor Root locus 2.

(2 p)

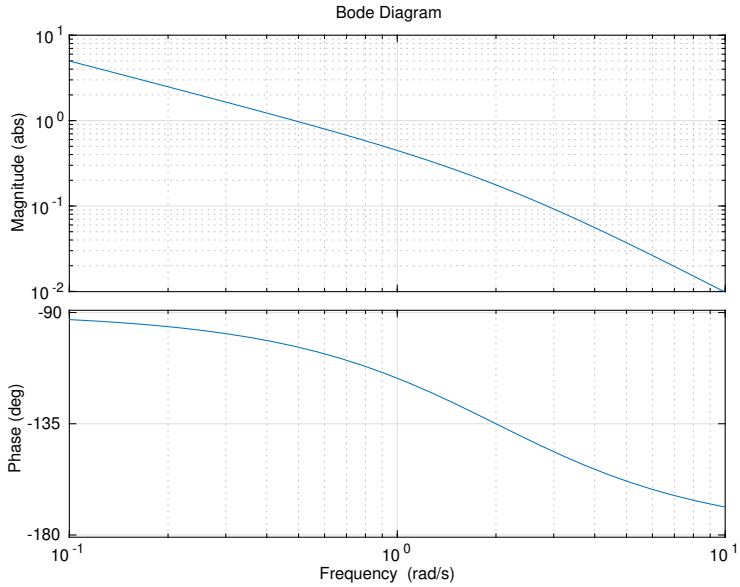


Figure 2: Bode diagram for $G(s)$ in Problem 3

Problem 3 (6/30)

Consider a system with input $u(t)$ and output $y(t)$ which is model by

$$Y(s) = G(s)U(s).$$

A bode plot for G is given in Figure 2.

a) Assume that the input $u(t) = 3\sin(3t)$. What will be the amplitude of the output $y(t)$ after all transients have faded?

(1 p)

b) We are interested in making the closed-loop system faster with a P-controller $F(s) = K$. Which value of K makes the crossover frequency of the open loop twice as fast as for G on its own?

(2 p)

c) What would be the phase margin for the open-loop system if we use the P-controller $F(s) = 2$.

(2 p)

d) Should you make the gain K in the P-controller smaller or larger than the value used in c) to reduce the stationary error? Motivate your answer.

(1 p)

Proposed solution to problem 3

a) From Figure 2 we have that $|G(3i)| \approx 0.1$, so the amplitude will be $3|G(3i)| \approx 0.3$.

(1 p)

b) The cross-over frequency for G is ≈ 0.5 rad/s. Our desired cross-over frequency is, hence, $\approx 2 \cdot 0.5 = 1$ rad/s. Morover, we havea that $|G(1i)| \approx 0.5$ With $K|G(1i)| \approx 1 \Leftrightarrow K \approx 2$.

(2 p)

c) With $F(s) = 2$ we will get a cross-over frequency of ≈ 1 rad/s. From the plot we have that $\arg G(1i) \approx 120°$, which corresponds to a phase margin of 60.

(2 p)

d) From the bode plot we see that $\|G(i\omega)\| \to \infty$ when $\omega \to 0$, which also means that $K\|G(i\omega)\| \to \infty$ when $\omega \to 0$ when $K \neq 0$. Hence, the stationary error will be zero, no matter ho K is selected.

(1 p)

Problem 4 (6/30)

You are given the state-space system

$$\begin{aligned}\dot{x}_1 &= x_1 - 2x_2 + u \\ \dot{x}_2 &= x_2 + u \\ y &= x_1\end{aligned}$$

a) Design a state-feedback controller $u = -Lx + \ell_0 r$ such that the closed-loop eigenvalues are in -2 and -3. Find ℓ_0 so that the steady-state tracking error is zero for $r(t) = \text{step}(t)$.

(3 p)

b) Design a state observer so that the state estimation error has both the eigenvalues in -10.

(3p)

Proposed solution to problem 4

a) First, we verify that the system is controllable:

$$\mathcal{S} = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \to \text{ok}$$

Then, we compute the desired characteristic polynomial

$$(\lambda + 2)(\lambda + 3) = \lambda^2 + 5\lambda + 6$$

The closed-loop characteristic polynomial is

$$\begin{aligned}
\varphi(\lambda) = \det(\lambda I - (A - BL)) &= \det \begin{bmatrix} \lambda - 1 + \ell_1 & 2 + \ell_2 \\ \ell_1 & \lambda - 1 + \ell_2 \end{bmatrix} \\
&= (\lambda - 1 + \ell_1)(\lambda - 1 + \ell_2) - \ell_1(2 + \ell_2) \\
&= \lambda^2 + (\ell_1 + \ell_2 - 2)\lambda + (-1 + \ell_1)(-1 + \ell_2) - \ell_1(2 + \ell_2) \\
&= \lambda^2 + (\ell_1 + \ell_2 - 2) + 1 - 3\ell_1 - \ell_2
\end{aligned}$$

Setting $\varphi(\lambda)$ to the desired characteristic polynomial we get

$$\begin{cases} \ell_1 + \ell_2 - 2 = 5 \\ 1 - 3\ell_1 - \ell_2 = 6 \end{cases} \to \begin{cases} \ell_2 = 7 - \ell_1 \\ \ell_1 = -6 \end{cases} \to L = \begin{bmatrix} -6 & 13 \end{bmatrix}$$

Finally, we need to compute ℓ_0 so that $G_c(0) = 1$.

$$\begin{aligned}
\ell_0 &= \frac{1}{C(-A + BL)^{-1}B} = \frac{1}{\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} -7 & 15 \\ -6 & 12 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix}} = \frac{6}{\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 12 & -15 \\ 6 & -7 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}} \\
&= -\frac{6}{\begin{bmatrix} 12 & -15 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}} = -\frac{6}{3} = -2
\end{aligned}$$

b) First, we verify that the system is observable:

$$\mathcal{S} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & -2 \end{bmatrix} \to \text{ok}$$

Then, we compute the desired characteristic polynomial

$$(\lambda + 10)(\lambda + 10) = \lambda^2 + 20\lambda + 100$$

The characteristic polynomial of the state estimation error is

$$\begin{aligned}
\varphi(\lambda) = \det(\lambda I - (A - KC)) &= \det \begin{bmatrix} \lambda - 1 + \kappa_1 & 2 \\ \kappa_2 & \lambda - 1 \end{bmatrix} \\
&= \lambda^2 + (\kappa_1 - 2)\lambda + 1 - \kappa_1 - 2\kappa_2
\end{aligned}$$

Matching the coefficients of this polynomial to those of the desired characteristic polynomial we get

$$\begin{cases} \kappa_1 - 2 = 20 \\ 1 - \kappa_1 - 2\kappa_2 = 100 \end{cases} \to \quad K = \begin{bmatrix} 22 \\ -60.5 \end{bmatrix}$$

Problem 5 (6/30)

ACME Inc is considering to expand its portfolios of products with drones. They hire you to design the control system of the propellers.

The minimum requirement is that the closed-loop bandwidth should be at least 40 rad/s.

a) You can choose what angular velocity sensor to use among three. Sensor A has a sampling frequency of 100 Hz, Sensor B has a sampling frequency of 250 Hz, and Sensor C has a sampling frequency of 1000 Hz. Which one works best for this application? Justify carefully your choice.

(1 p)

After some days of work, you come up with the continuous-time controller

$$F(s) = 2\frac{1+10s}{s(1+25s)} \tag{1}$$

which achieves a closed-loop bandwidth of 40 rad/s and a phase margin of 50°.

b) Discretize the controller with the sampling time chosen at point **(a)**, and write the discrete-time control law in the form

$$u_k = a_1 u_{k-1} + a_{k-2} u_{k-2} + b_0 e_k + b_1 e_{k-1} + b_2 e_{k-2}$$

(2.5 p)

c) Quantify the phase margin that is lost due to the discretization of the controller. How much is the remaining phase margin? Is it enough?

(1.5 p)

Proposed solution to problem 5

a) According to the "rule of thumb", the sampling frequency ω_s should be in the range $(20\omega_c, 50\omega_c)$, that is, $\omega_s \in (800, 2000)$ rad/s. Therefore, $f_s \in (127, 320)$ Hz. Sensor B is therefore the most appropriate.

b) The controller first need to be discretized, e.g. with Backward Euler. The sampling time is $T = \frac{1}{f_s} = \frac{1}{250} = 4 \cdot 10^{-3}$.

$$\begin{aligned} F(q) =& 2\frac{1 + 10\frac{1-q^{-1}}{4\cdot 10^{-3}}}{\frac{1-q^{-1}}{4\cdot 10^{-3}}\left(1 + 25\frac{1-q^{-1}}{4\cdot 10^{-3}}\right)} = 2\frac{10.004 - 10q^{-1}}{1 - q^{-1} + 6250(1 + q^{-2} - 2q^{-1})} \\ =& \frac{20.008 - 20q^{-1}}{6251 - 12501q^{-1} + 6250q^{-2}} \end{aligned}$$

The control law is therefore

$$6251u_k - 12501u_{k-1} + 6250u_{k-2} = 20.008e_k - 20e_{k-1}$$

$$\boxed{u_k = \frac{12501}{6251}u_{k-1} - \frac{6250}{6251}u_{k-2} + \frac{20.008}{6251}e_k - \frac{20}{6251}e_{k-1}}$$

c) The discretization causes a phase loss of

$$\Delta\varphi_m = -\frac{\omega_c}{2\pi f_s}180^\circ \approx -5^\circ$$

The residual phase margin is $50^\circ - 5^\circ \approx 45^\circ$ which is acceptable.